

AMAL MOHAN

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Backend-focused developer skilled in building full-stack, database-driven web applications with Python and Django. Experienced in authentication, authorization, role-based access control, designing PostgreSQL databases, managing source code with Git, and deploying Django applications. Currently expanding into AI development with TensorFlow, with a focus on clean code and responsive web design.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML5, CSS3

Frameworks: Django, Bootstrap

Database: PostgreSQL

Tools: Git, GitHub, VS Code, Render

Concepts: Object-Oriented Programming (OOP), CRUD Operations, Authentication & Authorization, Role-Based Access Control (RBAC), Database Design, Responsive Web Design, Deployment, Debugging

Currently Learning: Generative AI

EDUCATION

Bachelor of Technology, Computer Science & Engineering

Mohandas College of Engineering and Technology — APJ Abdul Kalam Technological University (KTU)

Graduation Year: 2026 | CGPA: 7.41

Relevant Coursework: Data Structures, Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

Diploma in Computer Science Engineering

NSS Polytechnic College, Pandalam

Completed: 2022 | CGPA: 8.7

PROJECTS

Academic Attendance Management System (AMS) | January 2025 – June 2026

Tech Stack: Python, Django, PostgreSQL, Bootstrap, JavaScript, Render

- Developed a full-stack academic attendance management system with separate Admin, Teacher, and Student portals supporting 3 distinct user roles.
- Implemented role-based authentication and authorization to control access across user roles.
- Built Department, Semester, Subject, Teacher, and Student management modules using a relational PostgreSQL database.
- Developed attendance tracking, attendance percentage calculation, and reporting/analytics functionality for teachers and admins.
- Integrated automated email notifications for attendance-related updates.
- Designed a responsive Bootstrap interface for desktop and mobile devices.
- Successfully tested with 10+ student records in the local development environment and deployed on Render with active administrator, teacher, and student accounts.

GitHub Repository: <https://github.com/AmalMohan487/Django-Attendance-System>

Live Demo: <https://django-attendance-system-gudk.onrender.com/>

AI Tomato Leaf Disease Detection System

Tech Stack: Python, Django, TensorFlow, MobileNetV2, OpenCV, Bootstrap

- Developed an AI-powered web application to detect tomato leaf diseases from uploaded images.
- Built an image upload and preprocessing pipeline using OpenCV.
- Used a CNN based on MobileNetV2 architecture for disease prediction.
- Developed the Django backend to process uploaded images and generate prediction results.
- Implemented disease explanations to help users understand prediction results.
- Developed a recommendation module allowing users to choose between Herbal and Chemical remedies based on the detected disease.
- Designed a responsive Bootstrap interface for image upload and result visualization.

GitHub Repository: <https://github.com/AmalMohan487/Agrosagemain>

CERTIFICATIONS

- Python Full-Stack Development — Zion IT Company (2023)

ACHIEVEMENTS

- Built and deployed a production-ready Academic Attendance Management System using Python, Django, PostgreSQL, and Render.
- Developed an AI-powered Tomato Leaf Disease Detection web application using TensorFlow, MobileNetV2, OpenCV, and Django.
- Designed and deployed a professional portfolio website showcasing technical projects and skills.
- Demonstrated continuous self-learning by expanding skills in Python, Django, backend development, and AI technologies.